

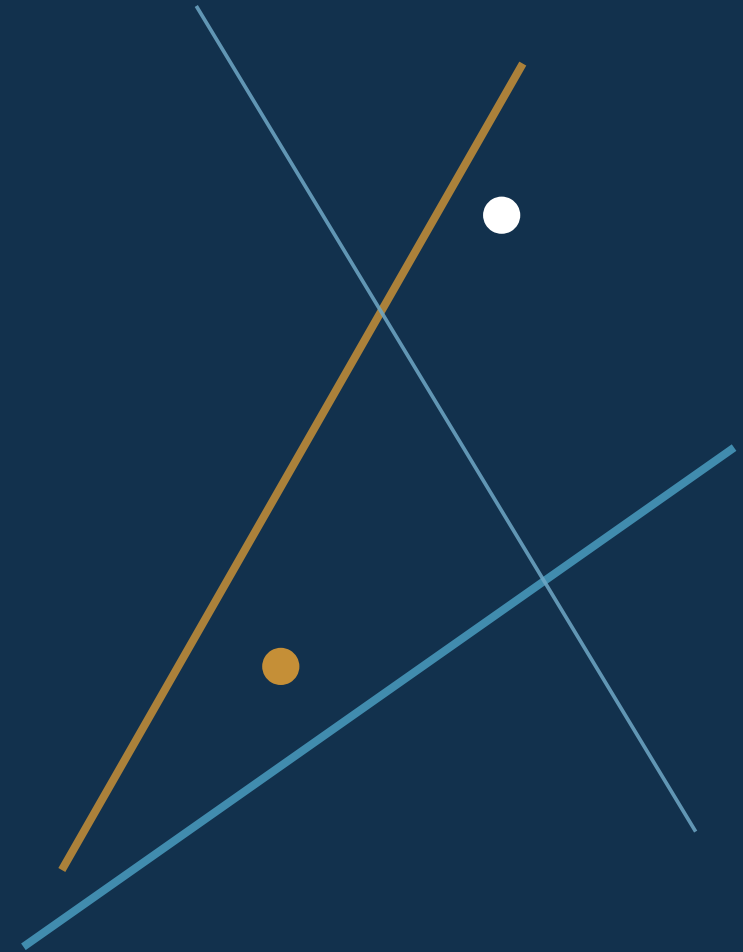
STAT 9120

# Linear Algebra Foundations

A bridge to the theory of linear models

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Fall 2026



# Learning objectives

What you should be able to see - not just compute

- Interpret matrices as linear transformations and collections of column vectors.
- Use span, rank, and null spaces to reason about identifiability in linear models.
- Understand least squares geometrically as orthogonal projection onto  $C(X)$ .
- Recognize why eigenvalues, quadratic forms, and generalized inverses appear in linear model theory.

**The organizing idea:  
 $Y = \text{projection} + \text{orthogonal residual}$**

$$Y = P_{C(X)}Y + \{I - P_{C(X)}\}Y$$

# Why linear algebra?

OVERVIEW

Nearly every theoretical question about the linear model is geometric

$$Y = X\beta + \varepsilon$$

## Existence

Does a least-squares solution exist?

## Uniqueness

Is  $\beta$  uniquely determined?

## Geometry

Why are fitted values projections?

## Inference

Why do quadratic forms generate sums of squares?

## PART I

# Vectors, spaces, and rank

Build the language needed to describe the model space  $C(X)$  and the directions that  $X$  cannot identify.

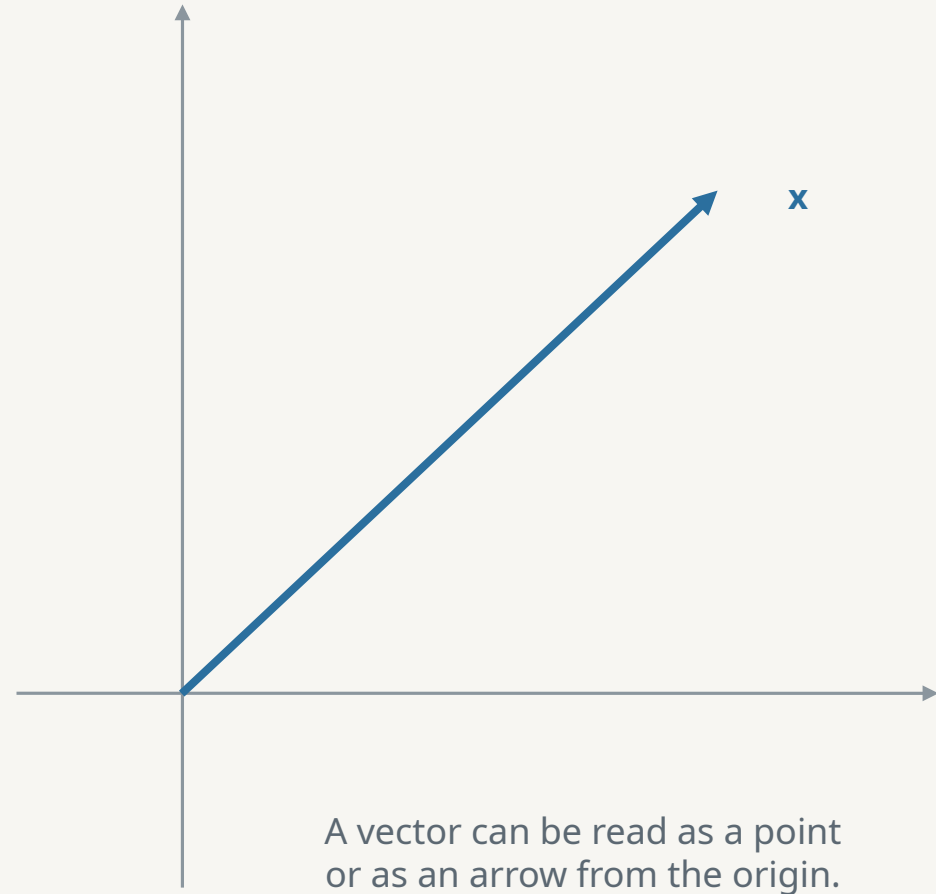
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# Vectors

A vector is an ordered point or direction in  $\mathbb{R}^n$

PART I

- Treat vectors as column vectors unless stated otherwise.
- Vector addition and scalar multiplication create new directions.
- In data analysis, an  $n$ -vector often stores one quantity for each observational unit.



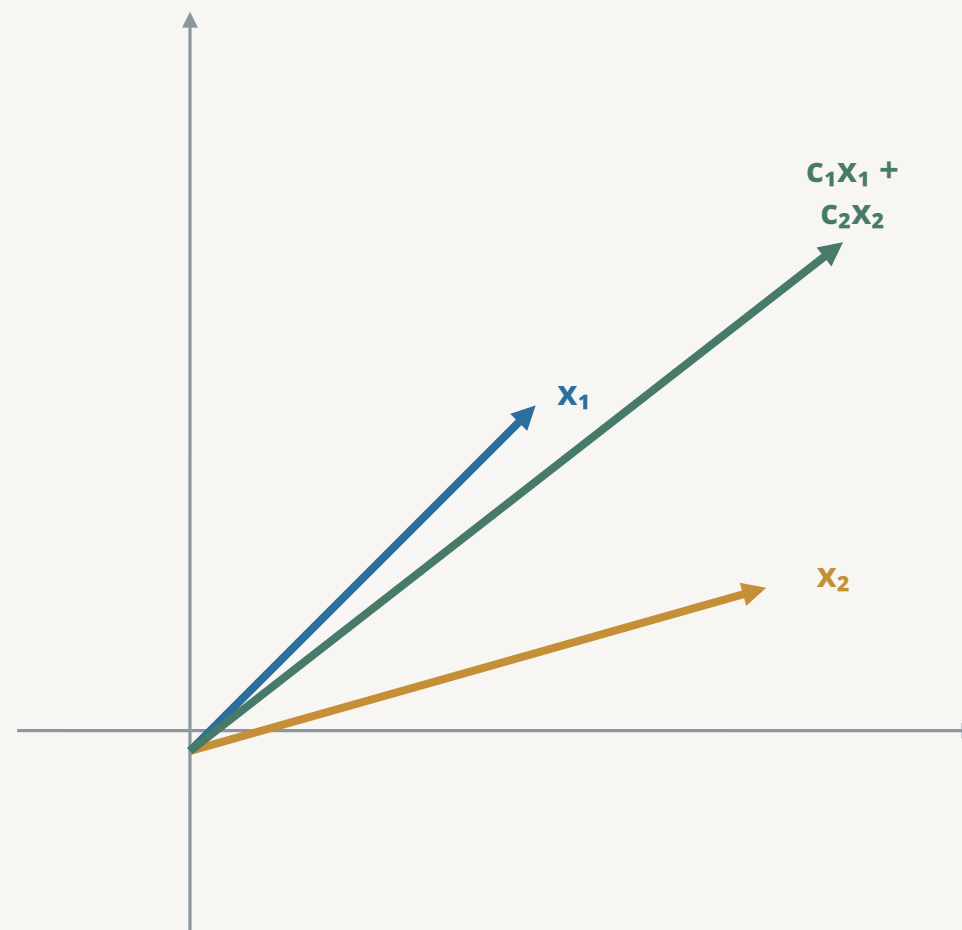
# Linear combinations

The basic operation behind matrix-vector multiplication

$$c_1x_1 + \cdots + c_px_p$$

## Example

$2x_1 - 3x_2$  is another vector built from  $x_1$  and  $x_2$ .



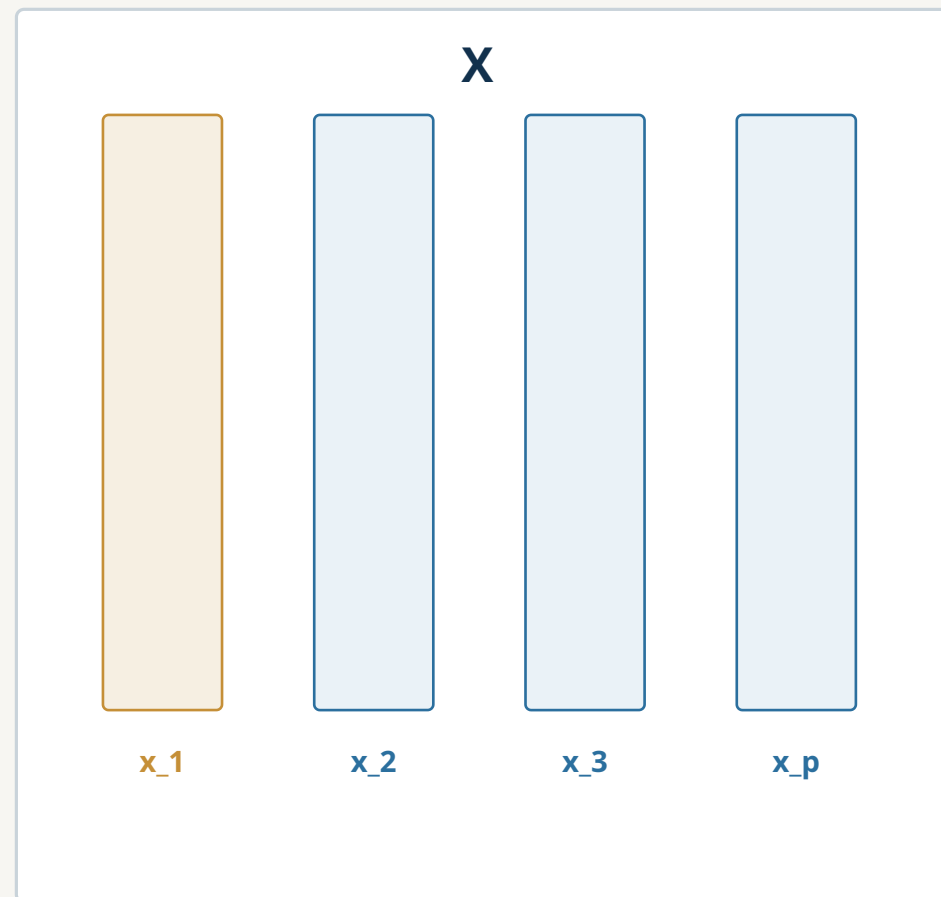
**Span = all possible linear combinations**

# Matrices as collections of columns

Column thinking is essential for linear models

$$A = (a_1 \quad a_2 \quad \cdots \quad a_p)$$

- $A \in \mathbb{R}^{n \times p}$  has  $n$  rows and  $p$  columns.
- Each column  $a_j$  is a vector in  $\mathbb{R}^n$ .
- The design matrix  $X$  stores the model components as columns.



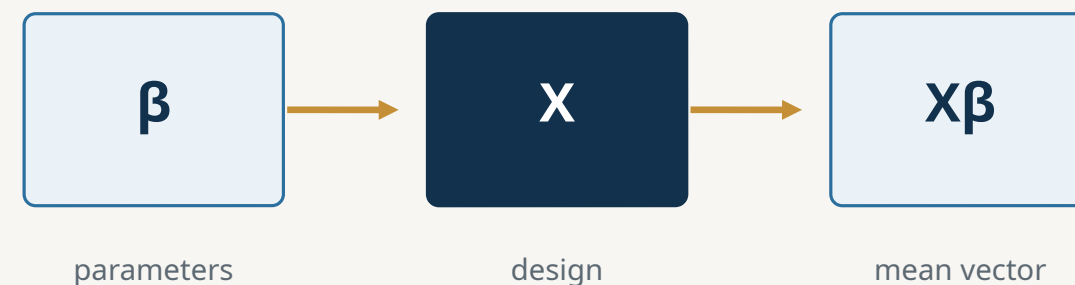
# Matrix-vector multiplication

$Ax$  is a linear combination of the columns of  $A$

$$Ax = x_1 a_1 + \cdots + x_p a_p$$

$$X\beta = \beta_1 x_1 + \cdots + \beta_p x_p$$

Therefore  $X\beta \in C(X)$



## Linear-model connection

The set of all possible mean vectors is exactly the column space  $C(X)$ .



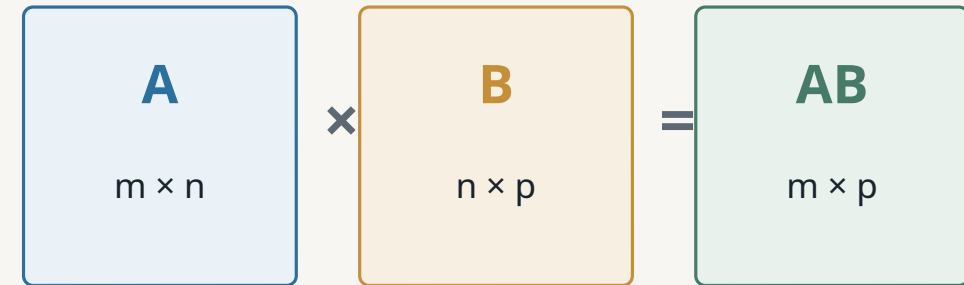
# Matrix multiplication

Always check the dimensions first

$$(m \times n)(n \times p) = (m \times p)$$

- The inner dimensions must agree.
- Matrix multiplication is associative:  $A(BC) = (AB)C$ .
- It is generally not commutative:  $AB \neq BA$ .

PART I



## Habit

Before doing any algebra, write the dimensions next to each matrix.

# Transpose and symmetry

Transpose reverses the order of products

$$(AB)^T = B^T A^T$$

$$A^T = A \implies A \text{ is symmetric}$$

- $X^T X$  and  $XX^T$  are always symmetric.
- Symmetric matrices have real eigenvalues and orthogonal eigenvectors.
- These properties make them central in linear model theory.

$$(X^T X)^T = X^T X$$

**Symmetry is not just cosmetic.**

It unlocks the spectral theorem and clean quadratic-form theory.

# Systems of linear equations

$Ax = b$  asks whether  $b$  can be built from the columns of  $A$

$$Ax = b$$

- Unique solution
- Infinitely many solutions
- No solution

## Unique

$b$  lies in  $C(A)$  and no nonzero direction is lost.

## Many

$b$  lies in  $C(A)$ , but  $A$  has a nontrivial null space.

## None

$b$  does not lie in  $C(A)$ .

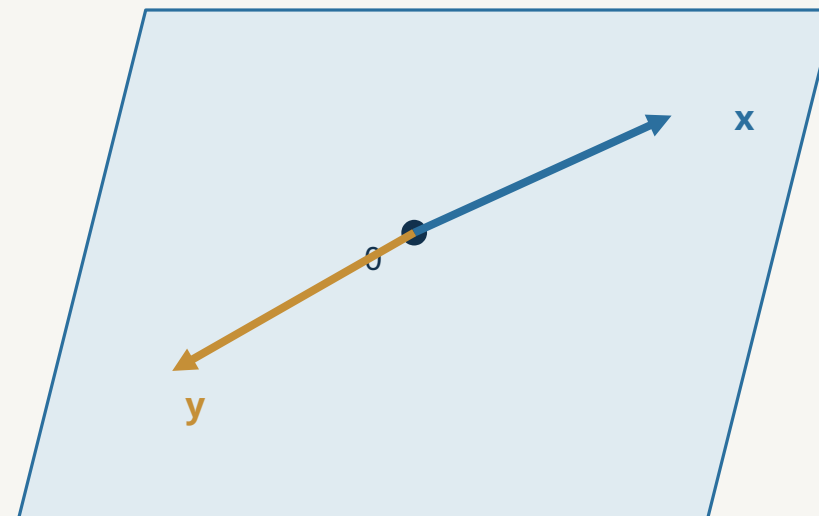
**$Ax = b$  is solvable  $\Leftrightarrow b \in C(A)$**

# Span and subspaces

Span is the set of everything your vectors can generate

$$\text{span}(x_1, \dots, x_p) = \left\{ \sum_{j=1}^p c_j x_j : c_j \in \mathbb{R} \right\}$$

- A subspace contains 0.
- It is closed under addition.
- It is closed under scalar multiplication.



**A subspace passes through the origin.**

# The column space

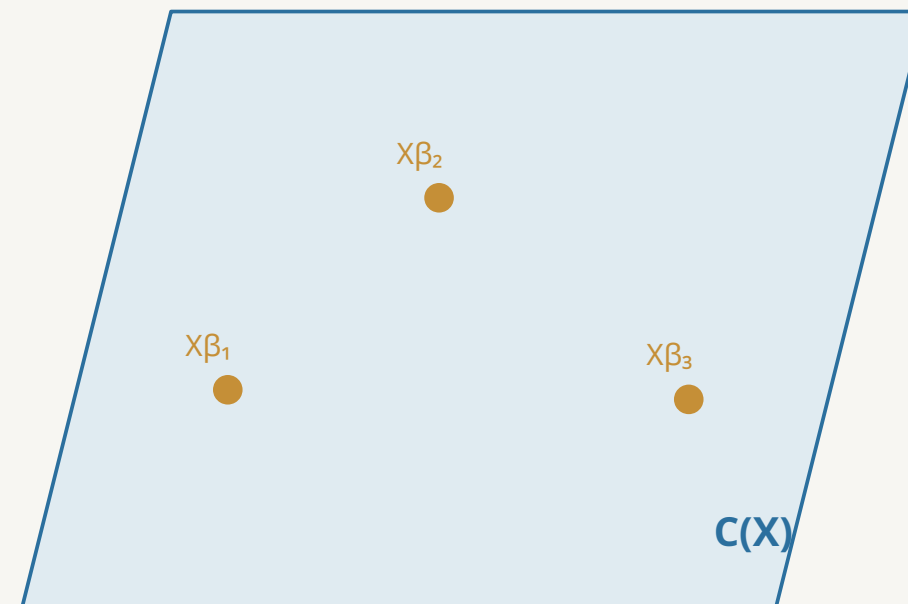
The model space of a linear model

$$\mathcal{C}(A) = \{Ax : x \in \mathbb{R}^p\}$$

$$E(Y) = X\beta \in \mathcal{C}(X)$$

## Geometric interpretation

A linear model restricts the mean vector  $E(Y)$  to lie in the subspace  $\mathcal{C}(X)$ .



Every possible mean vector lives on the same subspace.

# Linear independence

Independence means no vector is redundant

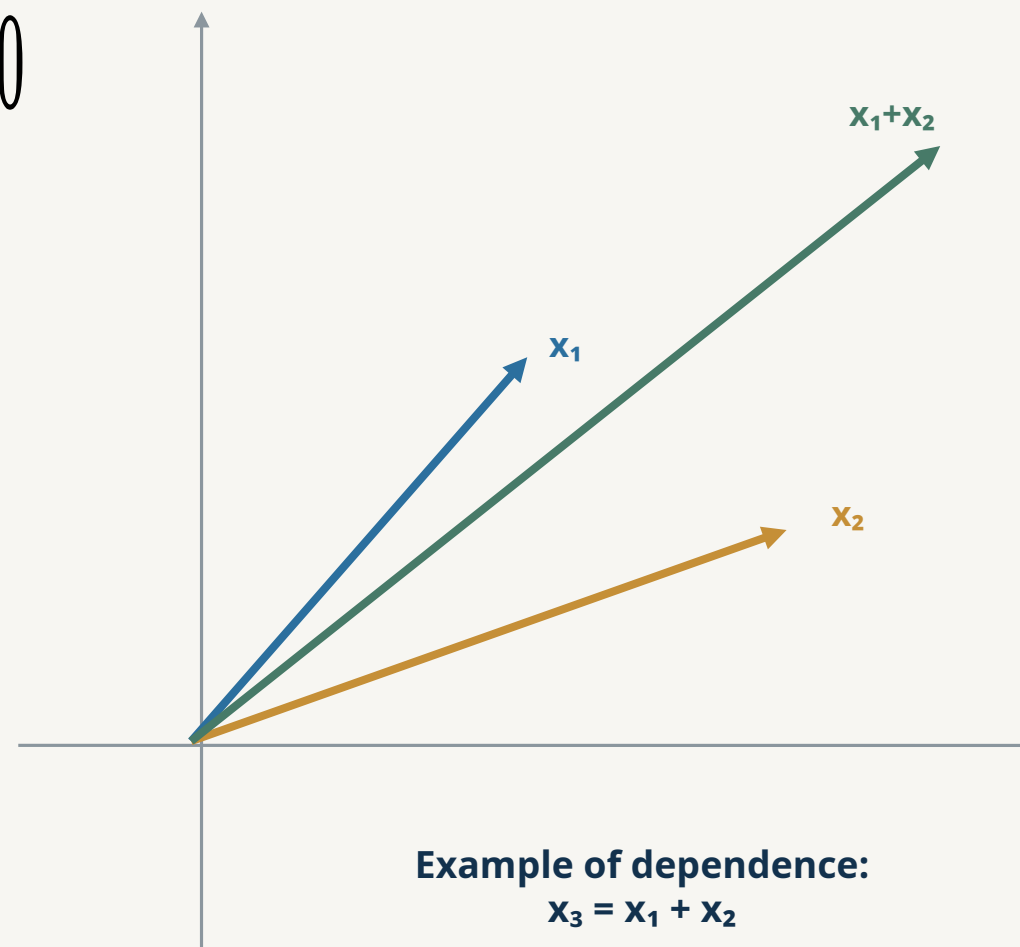
$$c_1x_1 + \cdots + c_px_p = 0 \implies c_1 = \cdots = c_p = 0$$

## Independent

No column can be reproduced from the others.

## Dependent

At least one nonzero coefficient vector  $c$  satisfies  $Xc = 0$ .

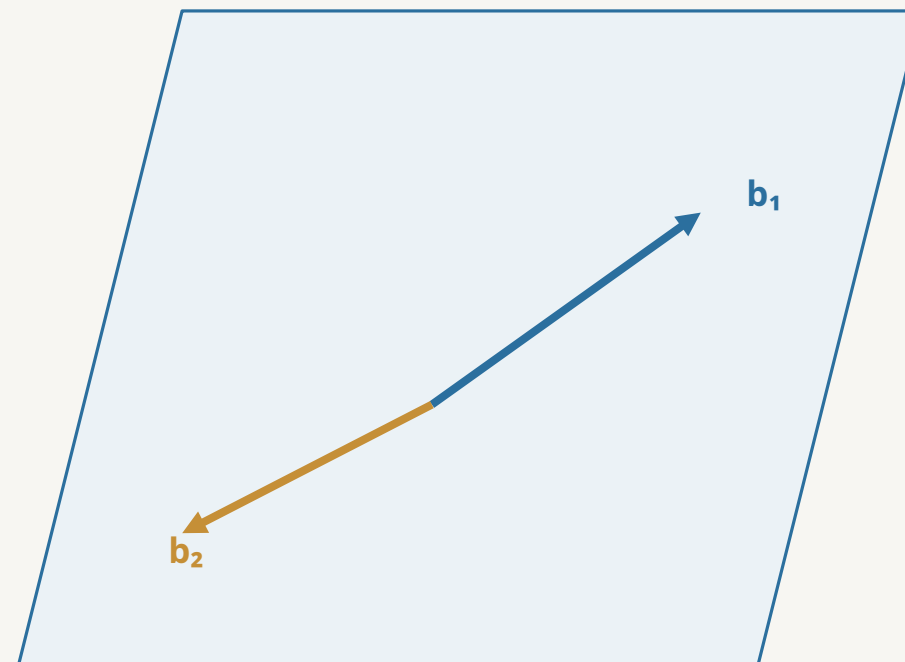


# Basis and dimension

A basis is a minimal set of directions that spans the space

- A basis spans the subspace.
- A basis is linearly independent.
- Every basis of the same subspace has the same number of vectors.

$$\dim(S) = \text{number of vectors in a basis for } S$$



**Two independent directions span this plane.**

# Rank

Rank counts the number of independent column directions

$$\text{rank}(A) = \dim \mathcal{C}(A)$$

- $\text{rank}(A) \leq \min(n, p)$ .
- Rank equals the maximum number of linearly independent columns.
- Column rank equals row rank.

## For a design matrix

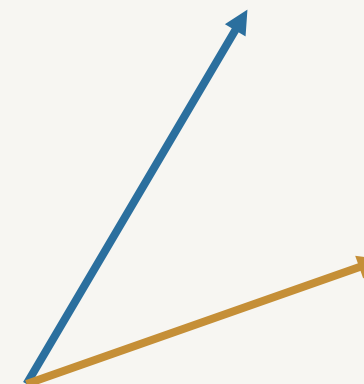
X has full column rank when  $\text{rank}(X) = p$ .

rank = 1



all columns lie on one line

rank = 2



two independent directions



# The null space

The directions that a matrix sends to zero

$$\mathcal{N}(A) = \{x : Ax = 0\}$$

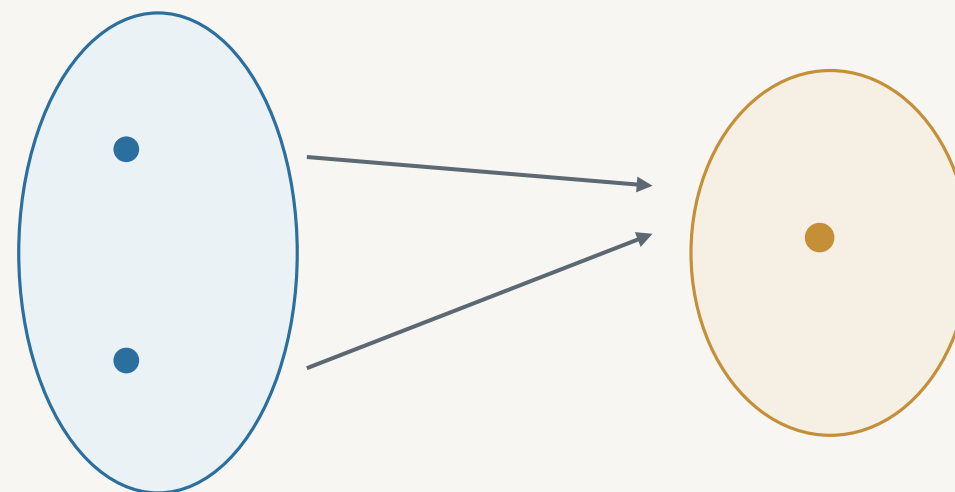
## Interpretation

If  $a \in \mathcal{N}(X)$ , moving  $\beta$  in direction  $a$  does not change  $X\beta$ .

$$X(\beta + a) = X\beta \quad \text{whenever } Xa = 0$$

parameter space

mean space



different  $\beta$  values can map  
to the same  $X\beta$

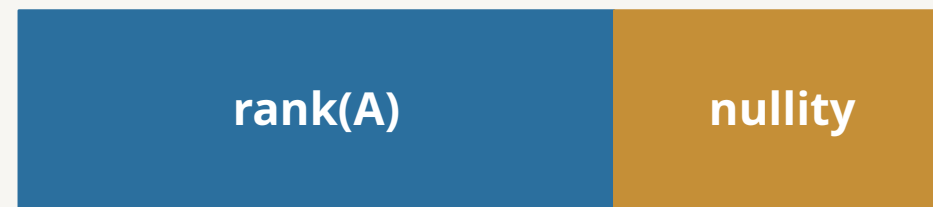
# Rank-nullity

Every parameter direction is either visible to A or lost by A

$$\text{rank}(A) + \dim \mathcal{N}(A) = p$$

- $\text{rank}(A)$ : directions that change  $Ax$
- $\text{nullity}(A)$ : directions that leave  $Ax$  unchanged

p-dimensional parameter space



## Linear-model meaning

Nonzero nullity means the parameterization has redundant directions.

# Identifiability in linear models

Column dependence translates directly into parameter nonuniqueness

$$Xa = 0, a \neq 0 \implies X(\beta + a) = X\beta$$

- Two different parameter vectors can produce the same mean vector.
- The data can identify  $X\beta$  even when it cannot identify  $\beta$  uniquely.
- This is the key distinction in less-than-full-rank models.

**Parameterization can be nonunique while the model space is unique.**

## Preview

Generalized inverses let us work with this situation without pretending  $X^T X$  is invertible.

# Full rank and $X^T X$

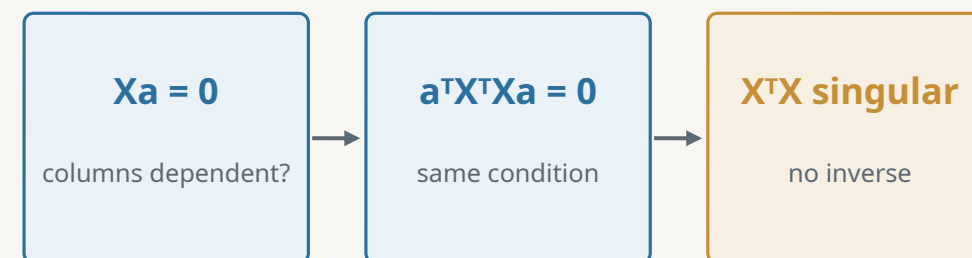
Why invertibility of  $X^T X$  is a rank question

$$a^T X^T X a = \|Xa\|^2$$

$$\mathcal{N}(X^T X) = \mathcal{N}(X)$$

**$\text{rank}(X) = p \Leftrightarrow X^T X$  is invertible**

- If  $Xa = 0$  only for  $a = 0$ , the columns of  $X$  are independent.
- Then  $X^T X$  is positive definite and has an ordinary inverse.



## Important proof pattern

Turn a matrix equation into a squared norm. Nonnegative quantities are easier to reason about.

# Part I recap

The algebraic vocabulary of the model space

PART I

$C(X)$

all possible mean vectors

$N(X)$

parameter directions that  $X$  cannot see

$\text{rank}(X)$

dimension of the model space

$\text{nullity}(X)$

dimension of parameter redundancy

**Next: geometry turns these spaces into least squares.**

# Orthogonality and projection

See ordinary least squares as geometry: decompose  $Y$  into a fitted component in  $C(X)$  and an orthogonal residual.

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# Inner products and norms

The algebra of length and angle

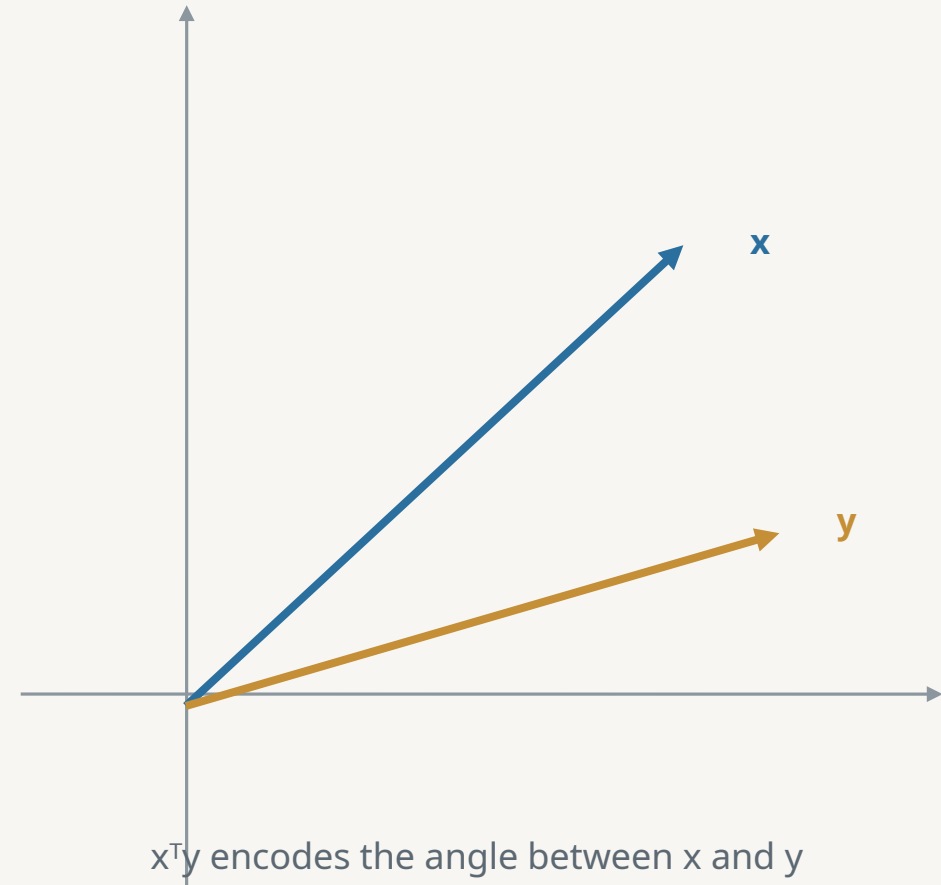
$$\langle x, y \rangle = x^T y$$

$$\|x\| = \sqrt{x^T x}$$

$$\|Y - X\beta\|^2 = (Y - X\beta)^T (Y - X\beta)$$

## Least-squares criterion

We choose  $\beta$  to minimize squared Euclidean distance between  $Y$  and  $X\beta$ .



# Orthogonality

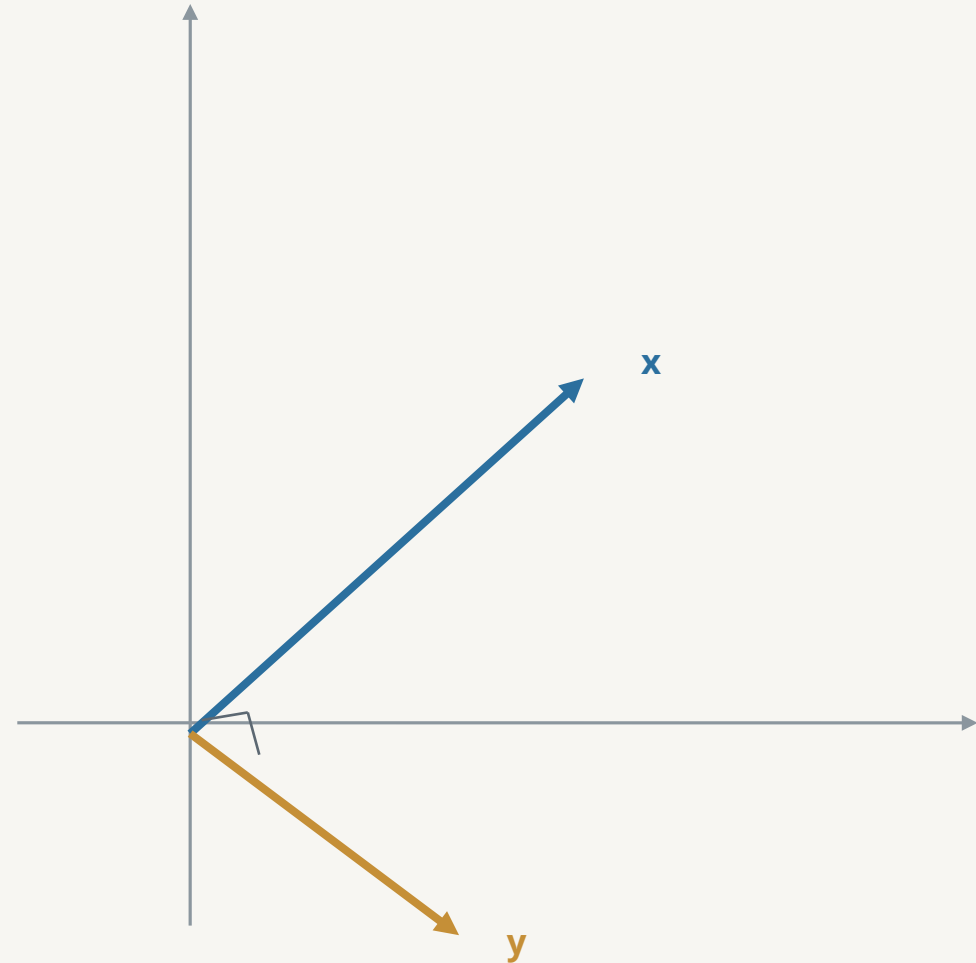
Perpendicular vectors have zero inner product

$$x \perp y \iff x^T y = 0$$

- Orthogonality is the linear-algebra version of a right angle.
- Residuals from least squares will be orthogonal to every column of  $X$ .

## Check

For  $x = (1,1)^T$  and  $y = (1,-1)^T$ ,  $x^T y = 0$ .





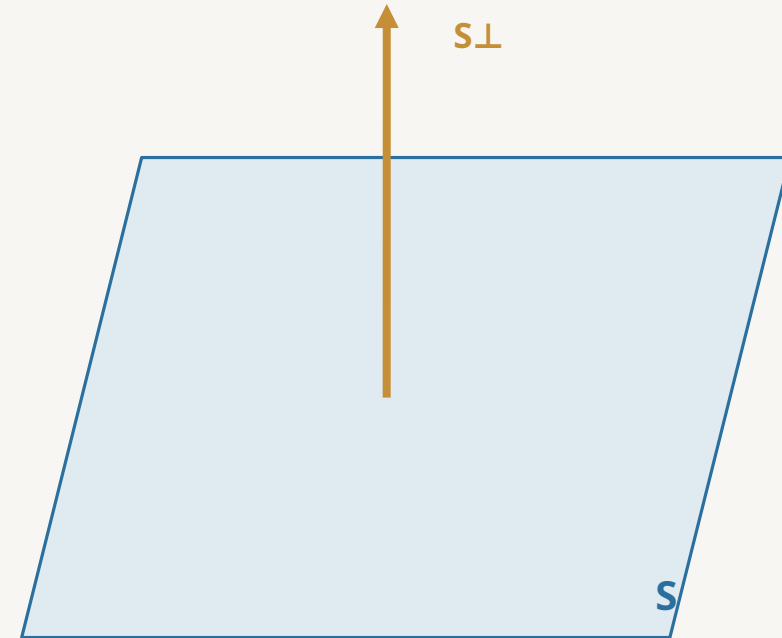
# Orthogonal complements

Every subspace comes with all directions perpendicular to it

$$S^\perp = \{y : y^T x = 0 \text{ for every } x \in S\}$$

$$\mathbb{R}^n = S \oplus S^\perp$$

- Every vector decomposes uniquely into one part in  $S$  and one part in  $S^\perp$ .
- This decomposition is exactly what least squares uses.



orthogonal directions leave the plane

# A fundamental orthogonality relation

Column space and null space are two sides of the same geometry

$$\mathcal{C}(X)^\perp = \mathcal{N}(X^T)$$

- $y \in \mathcal{C}(X)^\perp$  means  $y$  is orthogonal to every column of  $X$ .
- That is equivalent to  $X^T y = 0$ .
- This will imply  $X^T e = 0$  for least-squares residuals.

## Translation

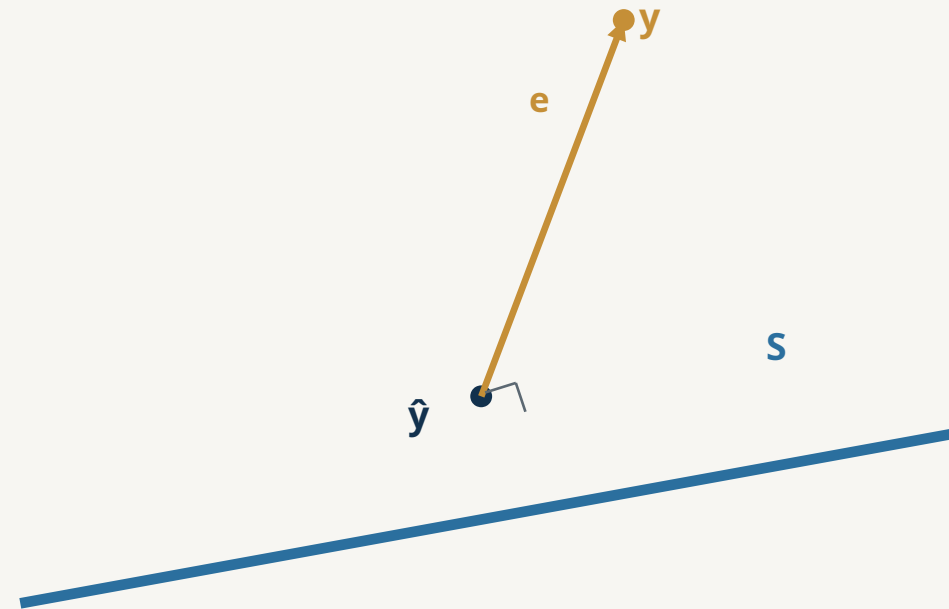
“orthogonal to every column” and “annihilated by  $X^T$ ” are the same statement.

# Orthogonal projection

The closest point in a subspace

$$P_S y = \arg \min_{z \in S} \|y - z\|^2$$

- $\hat{y} = P_S y$  lies in  $S$ .
- $e = y - \hat{y}$  lies in  $S^\perp$ .
- The decomposition  $y = \hat{y} + e$  is unique.



**Projection = nearest point + perpendicular residual**

# Least squares is projection

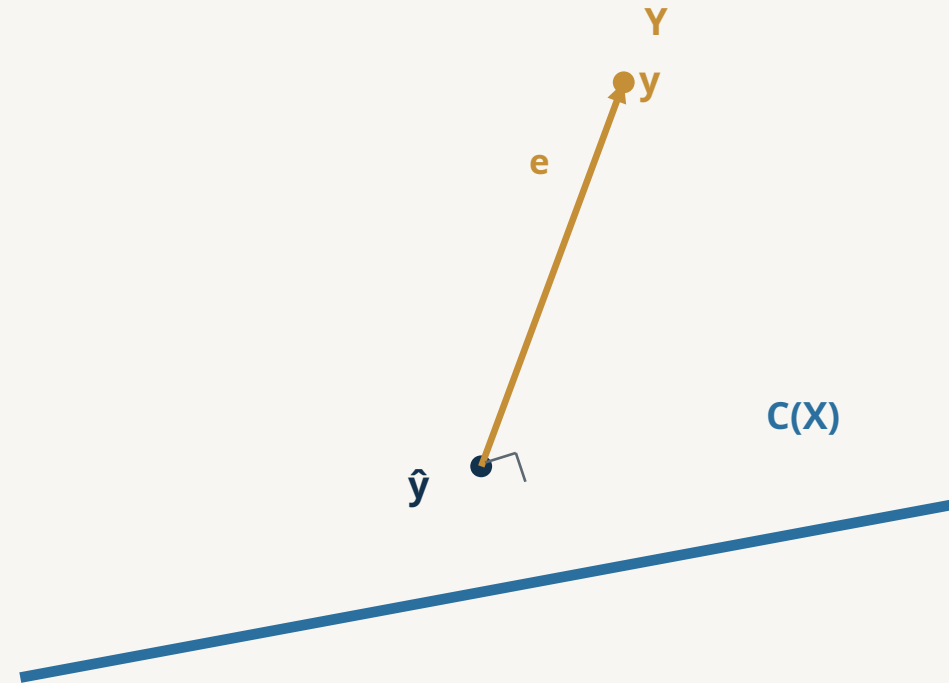
The fitted vector is the point in  $C(X)$  closest to  $Y$

$$\hat{Y} = P_{C(X)} Y$$

$$\hat{\beta} = \arg \min_{\beta} \|Y - X\beta\|^2$$

## Key geometry

Because every  $X\beta$  lies in  $C(X)$ , minimizing over  $\beta$  is the same as finding the closest point in  $C(X)$ .



# The normal equations

Orthogonality produces the least-squares equations

$$e = Y - X\hat{\beta}$$

$$e \perp \mathcal{C}(X) \implies X^T e = 0$$

$$X^T(Y - X\hat{\beta}) = 0$$

$$X^T X \hat{\beta} = X^T Y$$

When  $X$  has full column rank

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

The normal equations are not an arbitrary differentiation trick.  
They encode the orthogonality of the residual to the model space.

# The hat matrix

A matrix representation of projection onto  $C(X)$

$$H = X(X^T X)^{-1} X^T$$

$$\hat{Y} = HY$$

- H maps any response vector into  $C(X)$ .
- H depends only on X, not on Y.
- H is symmetric and idempotent.



## Projection identity

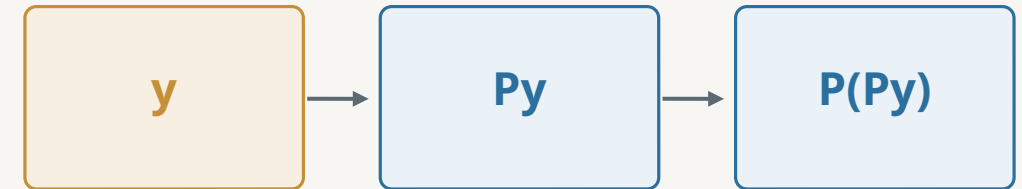
$HX = X$ : H leaves every vector already in  $C(X)$  unchanged.

# Projection matrices

Symmetry + idempotence characterize orthogonal projection

$$P^T = P \quad P^2 = P$$

- $P^2 = P$ : projecting twice does nothing new.
- $P^T = P$ : the projection is orthogonal rather than oblique.
- For  $H = X(X^T X)^{-1} X^T$ , both properties follow directly.



$$P(Py) = Py$$

## Useful theorem

If  $P$  is symmetric and idempotent, it is the orthogonal projection onto  $C(P)$ .

# The residual projection

$I - H$  projects onto  $C(X)^\perp$

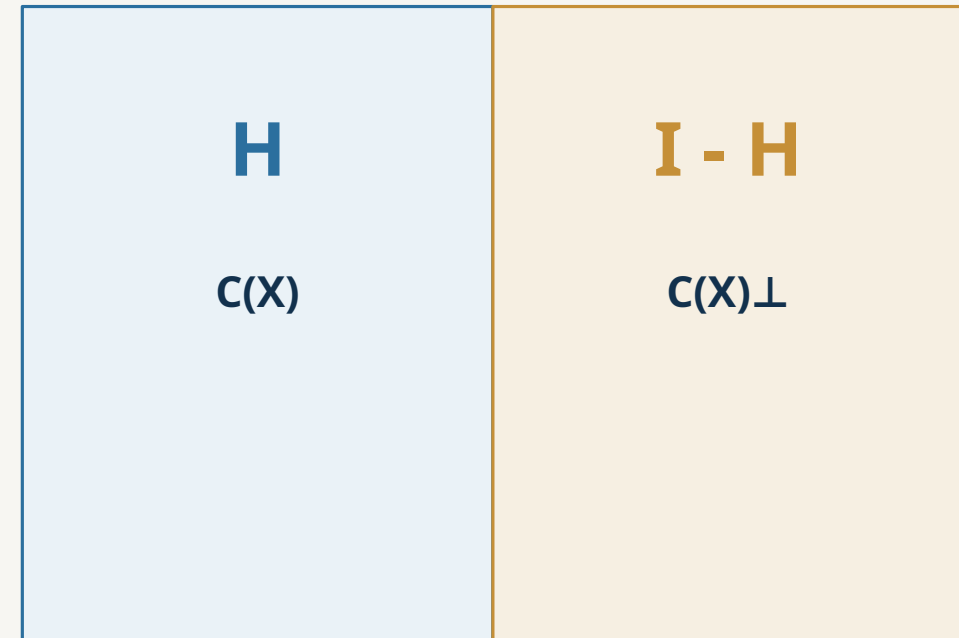
$$M = I - H$$

$$e = MY = (I - H)Y$$

$$(I - H)^2 = I - H$$

## Complementary projections

$H$  and  $I - H$  split  $\mathbb{R}^n$  into  $C(X)$  and  $C(X)^\perp$ .





# Pythagorean decomposition

Orthogonality turns geometry into sums of squares

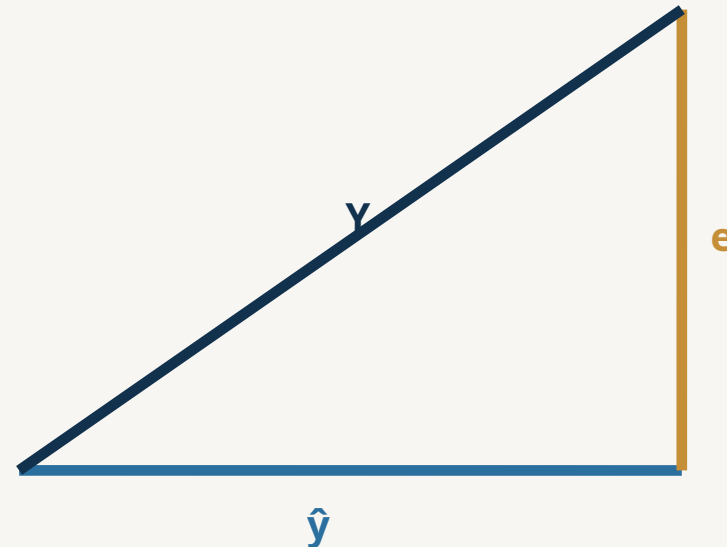
$$Y = \hat{Y} + e, \quad \hat{Y} \perp e$$

$$\|Y\|^2 = \|\hat{Y}\|^2 + \|e\|^2$$

$$Y^T Y = Y^T H Y + Y^T (I - H) Y$$

## ANOVA preview

Many sums-of-squares decompositions are Pythagorean identities in disguise.



# Part II recap

PART II

Least squares is an orthogonal decomposition

$$Y = HY + (I - H)Y$$

**HY**

fitted component in  $C(X)$

**(I-H)Y**

residual component in  $C(X)^\perp$

**$X^T e = 0$**

normal equations

**This geometry drives estimation, ANOVA, and many distributional results.**

# Eigenvalues, quadratic forms, and singular models

The machinery that supports distributions of sums of squares and less-than-full-rank linear models.

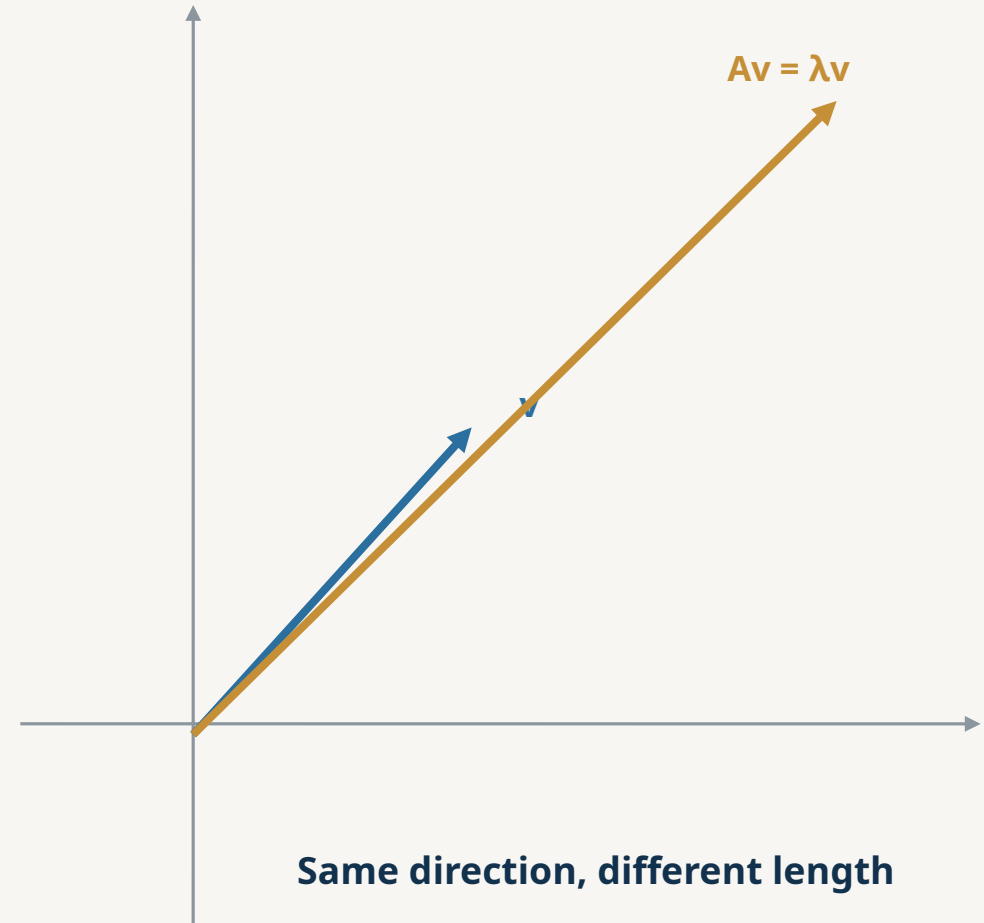
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# Eigenvalues and eigenvectors

Special directions preserved by a linear transformation

$$Av = \lambda v, \quad v \neq 0$$

- $v$  is an eigenvector.
- $\lambda$  is its eigenvalue.
- $A$  stretches, shrinks, or reverses  $v$  without changing its direction.

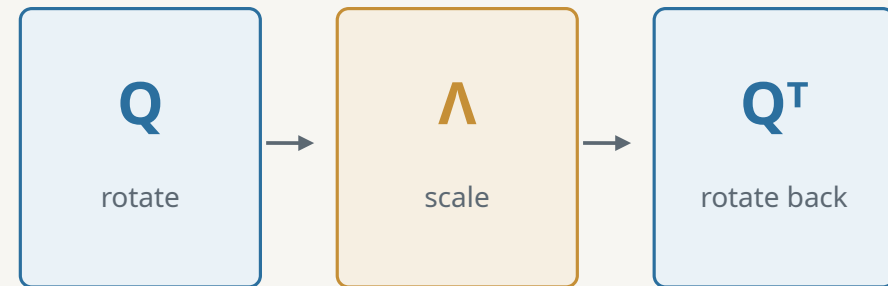


# The spectral theorem

Symmetric matrices admit an orthogonal eigenbasis

$$A = A^T \implies A = Q\Lambda Q^T$$

- All eigenvalues are real.
- Eigenvectors can be chosen orthonormal.
- $Q$  is orthogonal:  $Q^T Q = I$ .
- $\Lambda$  is diagonal and stores the eigenvalues.



## Why we care

Quadratic forms become simple weighted sums of squares in the eigenbasis.

# Projection eigenvalues, trace, and rank

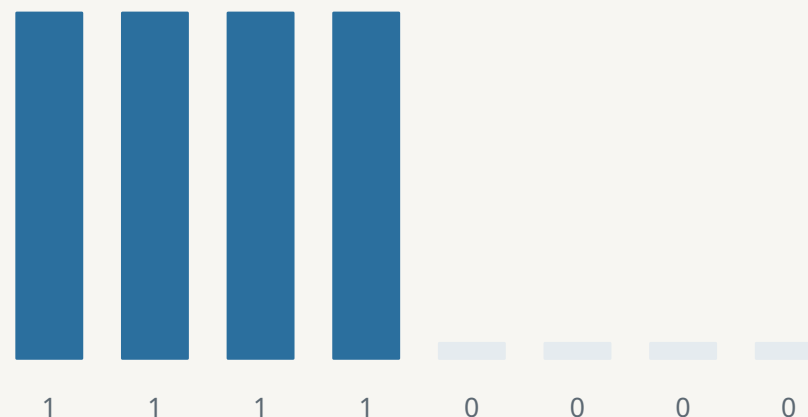
Idempotence forces a remarkably simple spectrum

$$P^2 = P, P v = \lambda v \implies \lambda^2 = \lambda$$

eigenvalues

Every eigenvalue of  $P$  is 0 or 1.

$$\text{tr}(P) = \sum_i \lambda_i = \text{rank}(P)$$



$\text{rank}(P) = \text{number of 1s}$

## Regression consequence

If  $\text{rank}(X)=p$ , then  $\text{tr}(H)=p$  and  $\text{rank}(I-H)=n-p$ .

# Quadratic forms

The matrix form of sums of squares

$$q(x) = x^T A x$$

$$SSE = Y^T (I - H) Y$$

$$\|Y - X\beta\|^2 = (Y - X\beta)^T (Y - X\beta)$$

$$\begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} a & b \\ b & c \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = ax_1^2 + 2bx_1x_2 + cx_2^2$$

## Why they matter

Under normality, distributions of quadratic forms generate chi-square and F results used throughout linear models.

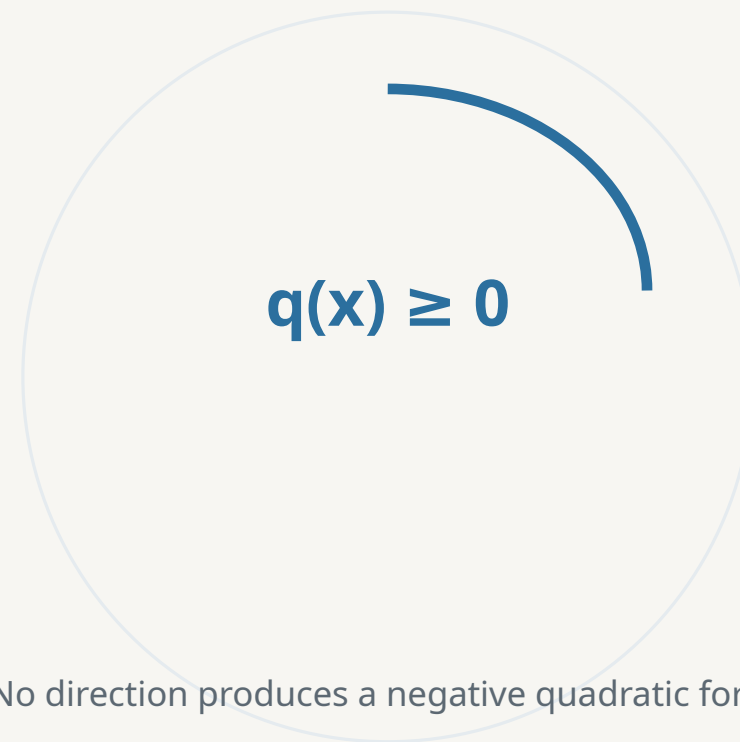
# Positive semidefinite matrices

Nonnegative quadratic forms

$$A \succeq 0 \iff x^T A x \geq 0 \text{ for all } x$$

$$A \succ 0 \iff x^T A x > 0 \text{ for all } x \neq 0$$

- For symmetric  $A$ , PSD means all eigenvalues are nonnegative.
- Positive definite means all eigenvalues are strictly positive.





# Why $X^T X$ is always positive semidefinite

A recurring one-line proof

$$a^T X^T X a = (Xa)^T (Xa) = \|Xa\|^2 \geq 0$$

**Therefore  $X^T X \geq 0$  for every  $X$ .**

## Full-rank refinement

$X^T X \succ 0$  exactly when  $X$  has full column rank.

- If  $Xa$  can equal 0 for some  $a \neq 0$ , the quadratic form can equal 0.
- If  $Xa = 0$  only at  $a = 0$ , the quadratic form is strictly positive away from the origin.

$$\text{rank}(X) = p \iff X^T X \succ 0$$

# Generalized inverses: a preview

What replaces  $(X^T X)^{-1}$  when  $X$  is rank deficient?

$$ABA = A$$

- $B$  is a generalized inverse of  $A$ .
- Generalized inverses need not be unique.
- They let us solve normal equations even when  $X^T X$  is singular.

$$B = (X^T X)^{-}$$

## Important invariant

Although  $B$  may not be unique,  $XB X^T$  is the unique orthogonal projection onto  $C(X)$ .

$$XB X^T = P_{C(X)}$$

**Nonunique parameters; unique fitted mean.**

# Worked example: a full-rank design

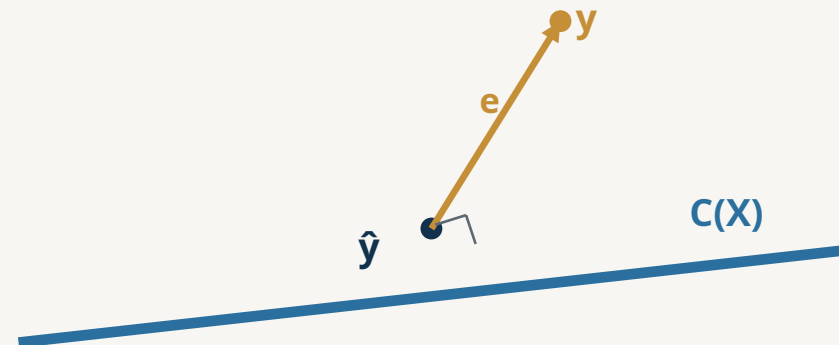
Simple linear regression with three observations

$$X = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix} \quad \text{rank}(X) = 2$$

$$X^T X = \begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix}$$

$$H = X(X^T X)^{-1} X^T$$

- $C(X)$  is a 2-dimensional plane in  $\mathbb{R}^3$ .
- $H$  projects any  $Y \in \mathbb{R}^3$  onto that plane.
- The residual lies along the 1-dimensional orthogonal complement.



# Worked example: a rank-deficient design

Redundant columns change the parameterization, not the model space

$$\mathbf{X} = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & 3 \end{pmatrix} \quad x_3 = x_1 + x_2$$

$$\mathbf{a} = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} \quad a \neq 0, \quad X\mathbf{a} = \mathbf{0}$$

$$X(\beta + c\mathbf{a}) = X\beta \quad \text{for every } c$$

## What remains unique

$C(X)$ , the projection of  $Y$  onto  $C(X)$ , and the fitted mean vector.

## What is not unique

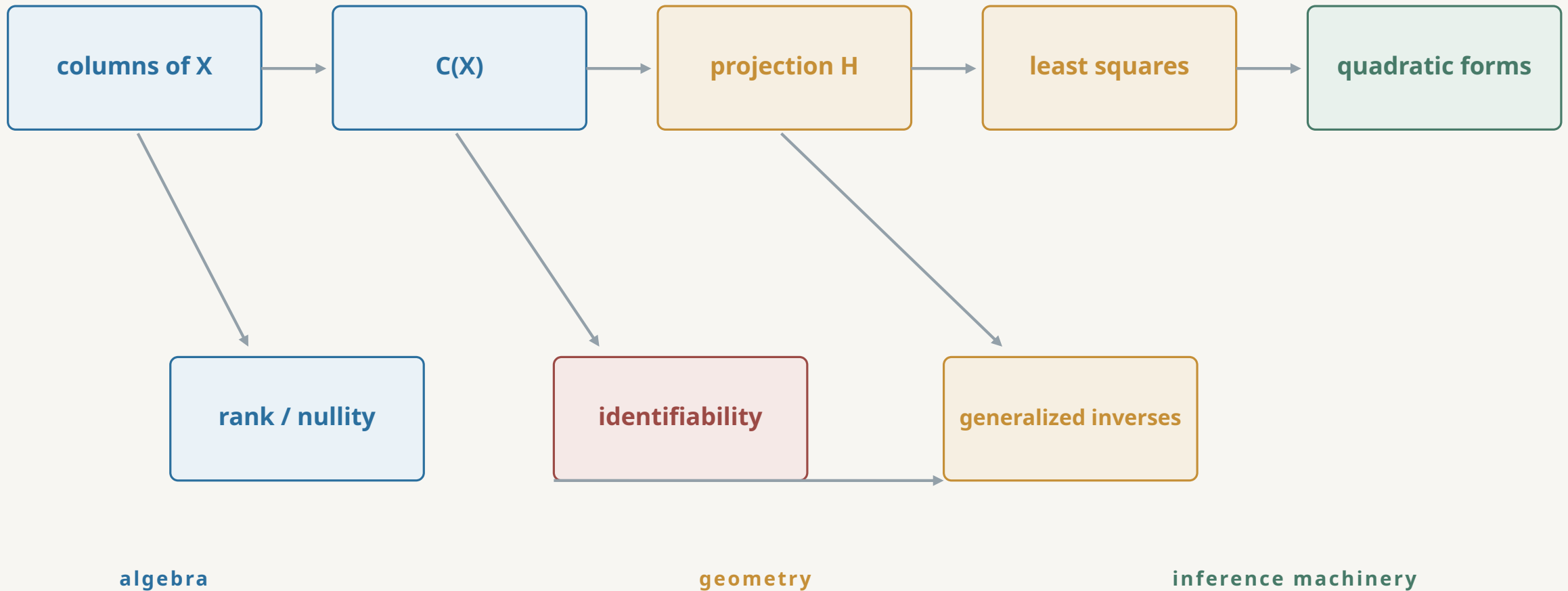
The coefficient vector  $\beta$  without additional constraints.

**The model is a subspace, not a particular coordinate system.**

# The big picture

How the pieces fit together

SUMMARY



# Practice problems

Use proofs to connect algebra and geometry

PRACTICE

- 1. Show  $N(X^T X) = N(X)$ .
- 2. If  $P^T = P$  and  $P^2 = P$ , show  $I - P$  is also symmetric and idempotent.
- 3. Show  $HX = X$  for  $H = X(X^T X)^{-1} X^T$ .
- 4. Show  $X^T(I - H) = 0$  and interpret it geometrically.
- 5. Show every eigenvalue of an idempotent matrix is 0 or 1.
- 6. For rank-deficient  $X$ , explain why  $\beta$  may be nonunique while  $X\beta$  is identifiable.

**For each proof: state the space, use dimensions, and interpret the result.**

# What to remember

$C(X)$  is the space of possible mean vectors.

$N(X)$  contains parameter directions that are not identifiable.

$H$  projects  $Y$  onto  $C(X)$ .

$I - H$  projects onto the residual space  $C(X)^\perp$ .

**Quadratic forms** encode the sums of squares used for inference.

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Next: full-rank linear models and the least-squares estimator